

Sharp factor-two excess transfer for commuting positive semidefinite matrices

Matthew J. Colbrook

Department of Applied Mathematics and Theoretical Physics
University of Cambridge, Cambridge, United Kingdom
m.colbrook@damtp.cam.ac.uk

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Verified partial result for RA-10. Theorem 1.1 and Section 3 give sharp factor-two transfer for a simultaneous eigenbasis and specified truncation projector. The general non-commuting nuclear question remains open. Independent agent verification is not external human peer review or formal certification. Original draft remarks about review or source access reflect the input date; the dated independent review records the subsequent checks.

Abstract

For every finite Schatten norm, we prove a sharp factor-two bound on relative approximation excess when the positive semidefinite input, its approximation, and the selected truncation projector have a common orthonormal eigenbasis. This is a partial result for the nuclear norm: it does not resolve the corresponding question for arbitrary noncommuting positive semidefinite matrices.

1 Statement

Let $f : [0, \infty) \rightarrow [0, \infty)$ be continuous and nondecreasing, with $f(x)/x$ nonincreasing on $(0, \infty)$. No assumption that $f(0) = 0$ is made. Let $A, B \succeq 0$ have a common orthonormal eigenbasis $q_1, \dots, q_n$, ordered so that the eigenvalues of A satisfy $a_1 \geq \dots \geq a_n \geq 0$. Fix $1 \leq k < n$ and a set $S \subseteq \{1, \dots, n\}$ of exactly k indices, and suppose $b_i = 0$ for $i \notin S$. Define

$$A = \sum_{i=1}^n a_i q_i q_i^T, \quad B = \sum_{i \in S} b_i q_i q_i^T, \quad P = \sum_{i \in S} q_i q_i^T, \quad C = \sum_{i \in S} f(b_i) q_i q_i^T.$$

In particular, the selected projector P also shares this eigenbasis. The definition of C includes $f(0)$ at the padded zero eigenvalues indexed by S ; it does not mean the full matrix $f(B)$ when $f(0) > 0$.

For $1 \leq p < \infty$, let $\|M\|_{S_p} = (\sum_i \sigma_i(M)^p)^{1/p}$ and put

$$T_A = \sum_{i>k} a_i^p, \quad T_f = \sum_{i>k} f(a_i)^p.$$

These are the optimal rank- k Schatten p -errors raised to the power p .

Theorem 1.1. *Suppose $\tau = a_{k+1} > 0$ and set $c = f(\tau)/\tau$. Then*

$$\|f(A) - C\|_{S_p}^p - T_f \leq 2c^p (\|A - B\|_{S_p}^p - T_A).$$

Consequently, for every $\eta \geq 0$,

$$\|A - B\|_{S_p}^p \leq (1 + \eta)T_A \implies \|f(A) - C\|_{S_p}^p \leq (1 + 2\eta)T_f.$$

For every $\varepsilon \geq 0$, the norm-relative implication is

$$\|A - B\|_{S_p} \leq (1 + \varepsilon)T_A^{1/p} \implies \|f(A) - C\|_{S_p} \leq (1 + 2\varepsilon)T_f^{1/p}.$$

For each fixed $p \in [1, \infty)$, both universal relative-excess constants two are optimal, even for operator-monotone power functions.

2 Proof

For $0 \leq x < y$ with $y \geq \tau$, subhomogeneity and continuity give

$$0 \leq f(y) - f(x) \leq \frac{f(y)}{y}(y - x) \leq c(y - x).$$

Also $f(x) \geq cx$ for $x \leq \tau$ and $f(x) \leq cx$ for $x \geq \tau$. If $f(\tau) = 0$, these inequalities and monotonicity force f to vanish identically, and the theorem is immediate.

Lemma 2.1. For all $a, b \geq 0$,

$$|f(a) - f(b)|^p \leq c^p(|a - b|^p + (\tau^p - a^p)_+).$$

Proof. If $\max(a, b) \geq \tau$, the pairwise Lipschitz inequality above suffices. If $0 \leq b \leq a \leq \tau$ and $a > 0$, subhomogeneity gives

$$f(a) - f(b) \leq f(\tau)(1 - b/a).$$

Writing $q = 1 - b/a \in [0, 1]$, we have

$$|a - b|^p + \tau^p - a^p - \tau^p q^p = (\tau^p - a^p)(1 - q^p) \geq 0,$$

which proves this case. If $0 \leq a \leq b \leq \tau$ and $b > 0$, then

$$f(b) - f(a) \leq f(\tau)(1 - a/b).$$

With $q = 1 - a/b \in [0, 1]$, the required inequality follows from

$$(\tau^p - b^p)q^p \leq \tau^p - b^p \leq \tau^p - a^p, \quad |a - b|^p = b^p q^p.$$

The case $a = b = 0$ is immediate; the other zero endpoints are included by continuity. $\square$

Proof of Theorem 1.1. Simultaneous diagonalization gives the exact decomposition

$$\delta_A := \|A - B\|_{S_p}^p - T_A = L_A + R_A,$$

where

$$L_A = \sum_{i \leq k} a_i^p - \sum_{i \in S} a_i^p, \quad R_A = \sum_{i \in S} |a_i - b_i|^p.$$

Define L_f, R_f by the analogous formulas with $f(a_i), f(b_i)$; then $\delta_f := \|f(A) - C\|_{S_p}^p - T_f = L_f + R_f$.

Match the selected indices outside $\{1, \dots, k\}$ with the omitted indices in $\{1, \dots, k\}$. Each matched pair consists of values $x \leq \tau \leq y$. Since $f(y) \leq cy$ and $f(x) \geq cx$,

$$0 \leq f(y)^p - f(x)^p \leq c^p(y^p - x^p).$$

Summing over these pairs gives $0 \leq L_f \leq c^p L_A$. Since every omitted y is at least τ , the same matching also gives

$$L_A \geq \sum_{i \in S} (\tau^p - a_i^p)_+.$$

Lemma 2.1 now yields

$$R_f \leq c^p (R_A + L_A), \quad \delta_f \leq c^p (R_A + 2L_A) \leq 2c^p \delta_A.$$

For the relative implications, use $c^p T_A \leq T_f$. Finally, convexity of $t \mapsto t^p$ gives

$$(1 + 2\varepsilon)^p \geq 2(1 + \varepsilon)^p - 1,$$

which converts the bound on the p th power into the norm-relative bound. $\square$

3 Sharpness and scope

Fix $p \in [1, \infty)$. Let $N \geq 1$, $0 < b < 1$, $0 < r < 1$, and take

$$A = I_{N+1} \oplus [0], \quad k = 1, \quad B = b e_{N+2} e_{N+2}^T, \quad f(x) = x^r.$$

Here $\tau = c = 1$, $T_A = T_f = N$, and

$$\|A - B\|_{S_p}^p = N + 1 + b^p, \quad \|f(A) - C\|_{S_p}^p = N + 1 + b^{pr}.$$

The ratio of p th-power excesses is

$$\frac{1 + b^{pr}}{1 + b^p}.$$

Taking $b = e^{-L}$ and $r = L^{-2}$ as $L \rightarrow \infty$ makes this ratio tend to two. With the simultaneous choice $N = L$ for positive integers L , the ratio of norm-relative excesses is

$$\frac{(1 + (1 + b^{pr})/N)^{1/p} - 1}{(1 + (1 + b^p)/N)^{1/p} - 1} \longrightarrow 2,$$

using $(1 + x/N)^{1/p} - 1 \sim x/(pN)$ for bounded positive x . For $p = 1$, the two ratios already coincide for every N .

These powers are operator monotone. Indeed, for $0 < r < 1$,

$$x^r = \frac{\sin(\pi r)}{\pi} \int_0^\infty t^{r-1} x(x+t)^{-1} dt.$$

Spectral calculus extends this identity to positive semidefinite matrices, and for $t > 0$ the map $X \mapsto X(X+tI)^{-1} = I - t(X+tI)^{-1}$ preserves their order. Integrating proves the assertion, including singular matrices.

If $\tau = 0$, the relative-error premise forces $A = B$. The selected space then contains range A , and $f(A) - C$ has exactly $n - k$ remaining eigenvalues equal to $f(0)$. Its Schatten p -error is the optimal tail. The relative conclusions therefore hold in this case too; no scaled inequality with an undefined quotient $f(0)/0$ is asserted.

For $p = 2$, a stronger result allowing noncommuting matrices is proved in the companion Frobenius manuscript. For $p = 1$, the present argument proves a sharp constant-two bound only with the specified simultaneous eigenbasis and selected projector. It does not settle existence of a universal constant for arbitrary noncommuting pairs. When this result is applied to a truncation $B = \hat{A}_k$, the selected projector must preserve the stated common basis, including within repeated eigenspaces and at padded zero eigenvalues.